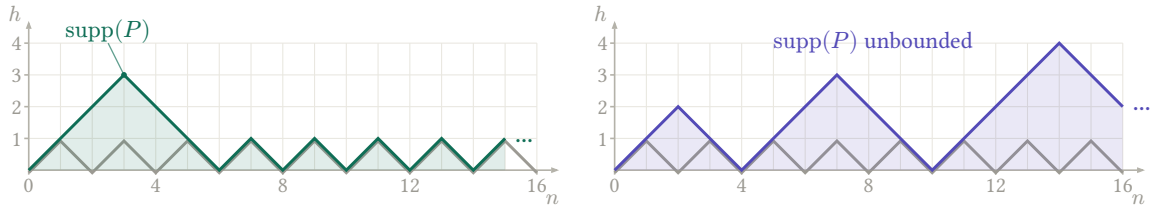
Dyck Paths with Infinite Support and Unbounded Temporal Logic

Introduction

> Instead of considering a finite total order, one can consider either an infinite (but discrete) total order or a more general partial order. In the former case, the combinatorial structure that is involved could be some notion of infinite Dyck path, that is, a Dyck path having infinite length. There are at least two different ways to formalize this suggestion: either working in the set of all Dyck paths (of any length), which is naturally partially ordered, and can be probably interpreted as some kind of infinitary limit object for the family of Dyck lattices; or introducing Dyck paths having infinite support, i.e., Dyck paths unbounded on the right which do not have a suffix consisting of the concatenation of an infinite number of peaks lying on the x-axis.
>
> — Ferrari

Both formalisations live inside the same picture. Read an infinite word over $\{u, d\}$ as a lattice path and compare it to the bottom element $\bot = (ud)^\omega$, the infinite concatenation of peaks lying on the axis: the two models are exactly the two ways a path can behave against $\bot$ in the long run.



Model 1: $P(ud)^\omega$, $|\mathrm{supp}(P)| < \infty$ Model 2: $|\mathrm{supp}(P)| = \infty$

Figure 1: The two shapes of infinite Dyck path, drawn against $\bot = (ud)^\omega$ (grey). On the left the path merges with $\bot$ and stays there, so it is a finite Dyck path padded with peaks; on the right its excursions above $\bot$ never stop.

Model 1 and 2

Setup

$$h_P(0) = 0, \qquad h_P(n) = \#\{i < n : p_i = u\} - \#\{i < n : p_i = d\}.$$

$$\mathcal{D}_\omega = \{P = p_0p_1p_2\cdots \in \{u,d\}^\omega \mid \forall n \in \mathbb{N}, h_P(n) \geq 0\}.$$

$$P \leq Q \quad \Leftrightarrow \quad \forall n \in \mathbb{N}, h_P(n) \leq h_Q(n).$$

$$\bot = (ud)^\omega, \qquad h_\bot(n) = \begin{cases} 0 \text{ if } & n = 0 \bmod 2 \\ 1 \text{ if } & n = 1 \bmod 2 \end{cases}.$$

$$\mathrm{supp}(P) = \{n \in \mathbb{N} \mid h_P(n) \neq h_\bot(n)\}, \qquad \mathrm{supp}_{\mathrm{at}}(P) = \{i \geq 1 \mid h_P(2i) > 0\}.$$

Lemma 1. $\mathrm{supp}_{\mathrm{at}}(P) = \{i \mid 2i \in \mathrm{supp}(P)\}$, *and* $|\mathrm{supp}(P)| < \infty \Leftrightarrow |\mathrm{supp}_{\mathrm{at}}(P)| < \infty$.

Proof. If $h_P(2i) = 0$ for all $i \geq m$, then each excursion between consecutive returns at distance 2 is a hill, so $h_P(2i+1) = 1 = h_\bot(2i+1)$ there as well; hence finiteness of $\mathrm{supp}_{\mathrm{at}}$ gives finiteness of supp. The converse is immediate. ∎

Lemma 2. $i \in \mathrm{supp}_{\mathrm{at}}(P) \Leftrightarrow \pi_i \leq P$, *where* $\pi_i = (ud)^{i-1}uudd(ud)^\omega$.

Lemma 3. *Pointwise* $\min$ *and* $\sup$ *of height functions of paths in* $\mathcal{D}_\omega$ *are height functions of paths in* $\mathcal{D}_\omega$*; hence* $(\mathcal{D}_\omega, \leq)$ *is a complete lattice with* $\bot = (ud)^\omega$, $\top = u^\omega$*, and binary meets distribute over arbitrary joins, so* $\mathcal{D}_\omega$ *is a complete Heyting algebra.*

Model 1

$$\iota_n : D_n \to D_{n+1}, \qquad \iota_n(P) = Pud.$$

Lemma 4. ι_n *is an order-embedding preserving* $\vee$, $\wedge$, *and* $\bot$*; it preserves neither* $\top$ *nor* $\sim$ *nor* $\rightsquigarrow$.

$$\mathcal{D}_{\text{colim}} = \operatorname*{colim}_n D_n, \qquad (n, P) \sim (m, Q) \quad \Leftrightarrow \quad \exists k \geq n, m : P(ud)^{k-n} = Q(ud)^{k-m},$$

$$[(n, P)] \leq [(m, Q)] \quad \Leftrightarrow \quad \exists k \geq n, m : P(ud)^{k-n} \underset{D_k}{\leq} Q(ud)^{k-m}.$$

$$j : \mathcal{D}_{\text{colim}} \hookrightarrow \mathcal{D}_\omega, \qquad j([(n, P)]) = P(ud)^\omega.$$

Proposition 5. j *is an order-embedding, and* $j(\mathcal{D}_{\text{colim}}) = \{P \in \mathcal{D}_\omega \mid |\text{supp}(P)| < \infty\}$.

Proof. ($\subseteq$) $j([(n, P)])$ agrees with $h_\bot$ from abscissa $2n$ on. ($\supseteq$) If $h_P = h_\bot$ from N on, then from N on the letters of P alternate ud, so $P = p(ud)^\omega$ with p a finite Dyck word. ∎

Model 2

$$\mathcal{D}_{\infty\text{-supp}} = \{P \in \mathcal{D}_\omega \mid |\text{supp}(P)| = \infty\}, \qquad \mathcal{D}_\omega = j(\mathcal{D}_{\text{colim}}) \sqcup \mathcal{D}_{\infty\text{-supp}}.$$

Proposition 6. $\mathcal{D}_{\infty\text{-supp}}$ *is an up-set of* $\mathcal{D}_\omega$*, closed under arbitrary joins, but closed under neither binary meets (interleaved combs:* P, Q *with pyramids at odd resp. even slots have* $P \wedge Q = \bot$*) nor pseudonegation; dually,* $j(\mathcal{D}_{\text{colim}})$ *is closed under* $\vee, \wedge, \bot$ *but not under* $\sim$, $\rightsquigarrow$*, or directed sups* ($\bigvee_n u^n d^n (ud)^\omega = u^\omega$).

$$R(P) = \{i \geq 1 \mid h_P(2i) = 0\}, \qquad \mathcal{D}_{\infty\text{-supp}} = \underbrace{\{|R(P)| = \infty\}}_{\text{recurrent}} \sqcup \underbrace{\{|R(P)| < \infty\}}_{\text{transient}}.$$

Logic

$$\mathcal{D}_\omega := \left\{P \in \{u, d\}^\omega \;\middle|\; \forall n \in \mathbb{N}, h_{P(n)} \geq 0\right\}.$$

$$\bot := (ud)^\omega.$$

$$\text{supp}(P) := \left\{n \in \mathbb{N} \;\middle|\; h_{P(n)} \neq h_{\bot(n)}\right\}.$$

$$\mathcal{D}_{\text{fs}} := \{P \in \mathcal{D}_\omega \mid |\text{supp}(P)| < \infty\}.$$

$$\mathcal{D}_{\text{is}} := \{P \in \mathcal{D}_\omega \mid |\text{supp}(P)| = \infty\}.$$

$$\mathcal{D}_\omega = \mathcal{D}_{\text{fs}} \sqcup \mathcal{D}_{\text{is}}.$$

Logic of Model 1

$$D_0 \to D_1 \to D_2 \to \cdots, \qquad \iota_{n(P)} = Pud.$$

$$\mathcal{D}_1 := \operatorname*{colim}_n D_n.$$

$$\mathcal{D}_1 \cong \mathcal{D}_{\text{fs}} = \{P \in \mathcal{D}_\omega \mid |\text{supp}(P)| < \infty\}.$$

$$\bot = (ud)^\omega \in \mathcal{D}_1.$$

$$\top \notin \mathcal{D}_1.$$

$$P, Q \in \mathcal{D}_1 \quad \Rightarrow \quad P \wedge Q \in \mathcal{D}_1, \qquad P \vee Q \in \mathcal{D}_1.$$

$$h_{P \wedge Q}(n) = \min\Big(h_{P(n)}, h_{Q(n)}\Big).$$

$$h_{P \vee Q}(n) = \max\Big(h_{P(n)}, h_{Q(n)}\Big).$$

$$\mathrm{supp}(P \wedge Q) = \mathrm{supp}(P) \cap \mathrm{supp}(Q).$$

$$\mathrm{supp}(P \vee Q) = \mathrm{supp}(P) \cup \mathrm{supp}(Q).$$

$$\mathcal{D}_1 = \text{distributive lattice with bottom, without top.}$$

$$\neg\exists\top \quad \Rightarrow \quad \neg\exists \text{ global Heyting implication.}$$

$$\mathcal{L}_1 = \{\bot, \wedge, \vee\}.$$

Logic of Model 2

$$\mathcal{D}_2 \coloneqq \mathcal{D}_{\mathrm{is}} = \{P \in \mathcal{D}_\omega \mid |\mathrm{supp}(P)| = \infty\}.$$

$$(ud)^\omega \notin \mathcal{D}_2.$$

$$\top = u^\omega \in \mathcal{D}_2.$$

$$P, Q \in \mathcal{D}_2 \quad \Rightarrow \quad P \vee Q \in \mathcal{D}_2.$$

$$\mathrm{supp}(P \vee Q) = \mathrm{supp}(P) \cup \mathrm{supp}(Q).$$

$$|\mathrm{supp}(P)| = \infty \quad \Rightarrow \quad |\mathrm{supp}(P \vee Q)| = \infty.$$

$$P, Q \in \mathcal{D}_2 \quad \neg\Rightarrow \quad P \wedge Q \in \mathcal{D}_2.$$

$$\mathrm{supp}(P \wedge Q) = \mathrm{supp}(P) \cap \mathrm{supp}(Q).$$

$$|\mathrm{supp}(P)| = \infty, \quad |\mathrm{supp}(Q)| = \infty \quad \neg\Rightarrow \quad |\mathrm{supp}(P) \cap \mathrm{supp}(Q)| = \infty.$$

$$\bot \notin \mathcal{D}_2, \qquad \wedge \text{ is not closed.}$$

$$\mathcal{D}_2 = \text{join-semilattice with top.}$$

$$\mathcal{L}_2 = \{\top, \vee\}.$$

$$\neg\exists\bot, \qquad \neg\exists \text{ global } \wedge, \qquad \neg\exists \text{ global } \to, \qquad \neg\exists \text{ global negation.}$$

Completion

$$\mathcal{D}_\omega = \mathcal{D}_{\mathrm{fs}} \sqcup \mathcal{D}_{\mathrm{is}}.$$

$$\mathcal{D}_1 \to \mathcal{D}_\omega \leftarrow \mathcal{D}_2.$$

$$\mathcal{D}_1 \cong \mathcal{D}_{\mathrm{fs}} \subseteq \mathcal{D}_\omega.$$

$$\mathcal{D}_2 = \mathcal{D}_{\mathrm{is}} \subseteq \mathcal{D}_\omega.$$

$$\bot = (ud)^\omega.$$

$$\top = u^\omega.$$

$$h_{\sup_{i\in I} P_i}(n) = \sup_{i\in I} h_{P_i}(n).$$

$$h_{\inf_{i\in I} P_i}(n) = \inf_{i\in I} h_{P_i}(n).$$

$$P \wedge \sup_{i\in I} Q_i = \sup_{i\in I} P \wedge Q_i.$$

$$\mathcal{D}_\omega = \text{frame} = \text{complete Heyting algebra}.$$

$$P \to Q = \sup\{R \in \mathcal{D}_\omega \mid P \wedge R \leq Q\}.$$

$$\neg P = P \to \bot.$$

$$\mathcal{L}_\omega = \{\bot, \top, \wedge, \vee, \to, \neg\}.$$

$$\mathcal{L}_\omega + \left\{\sup_{i\in I}, \inf_{i\in I}\right\}.$$

$$\mathrm{Idl}(\mathcal{D}_1) \cong \mathcal{D}_\omega.$$

Final overview

Property	Model 1	Model 2	Completion
Space	$\mathcal{D}_1 = \operatorname{colim}_n D_n$	$\mathcal{D}_2 = \mathcal{D}_{\text{is}}$	$\mathcal{D}_\omega$
Support	$\lvert\text{supp}(P)\rvert < \infty$	$\lvert\text{supp}(P)\rvert = \infty$	$\lvert\text{supp}(P)\rvert \leq \infty$
Bottom	$(ud)^\omega$	$\neg\exists$	$(ud)^\omega$
Top	$\neg\exists$	u^ω	u^ω
$\wedge$	✓	×	✓
$\vee$	✓	✓	✓
$\to$	×	×	✓
$\neg$	×	×	✓
Arbitrary meets	×	×	✓
Arbitrary joins	×	nonempty	✓
Algebra	distributive lattice + bottom	join-semilattice + top	complete Heyting algebra / frame
Logic	$\bot, \wedge, \vee$	$\top, \vee$	$\bot, \top, \wedge, \vee, \to, \neg$
Relation	$\mathcal{D}_1 \cong \mathcal{D}_{\text{fs}} \subseteq \mathcal{D}_\omega$	$\mathcal{D}_2 = \mathcal{D}_{\text{is}} \subseteq \mathcal{D}_\omega$	$\mathcal{D}_\omega = \mathcal{D}_{\text{fs}} \sqcup \mathcal{D}_{\text{is}}$

Bibliography